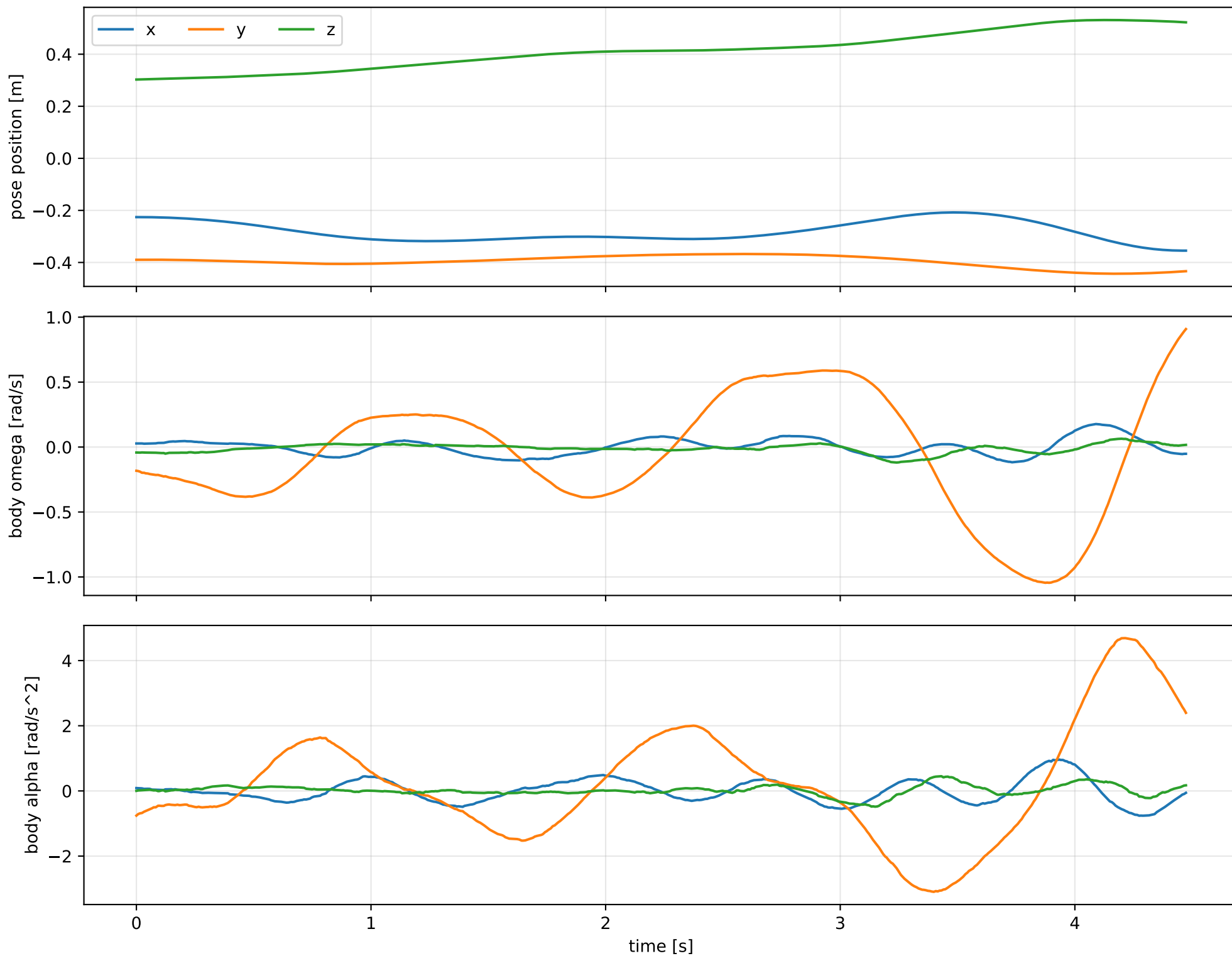
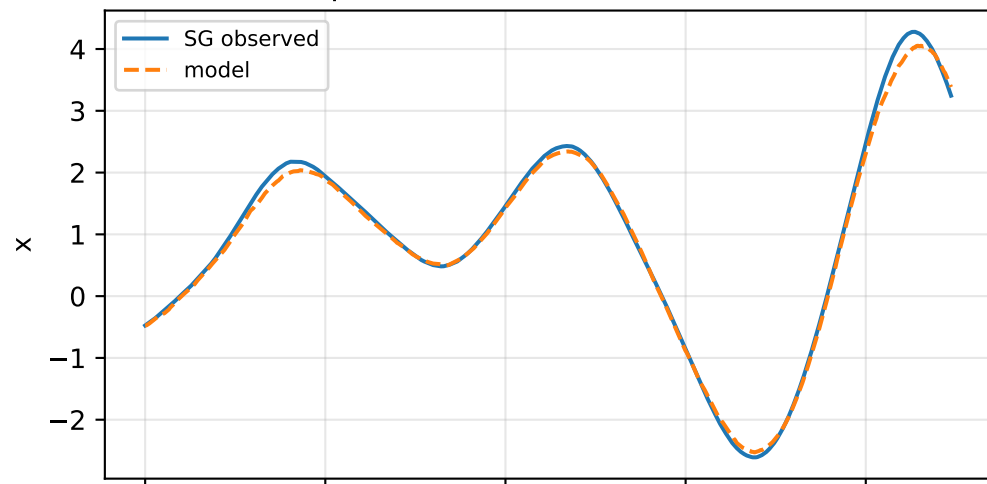
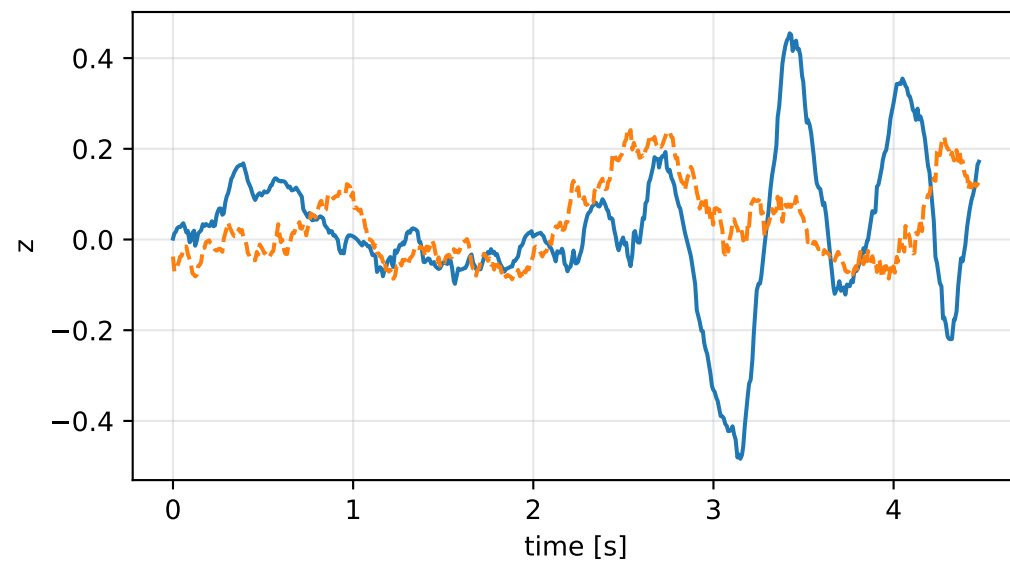
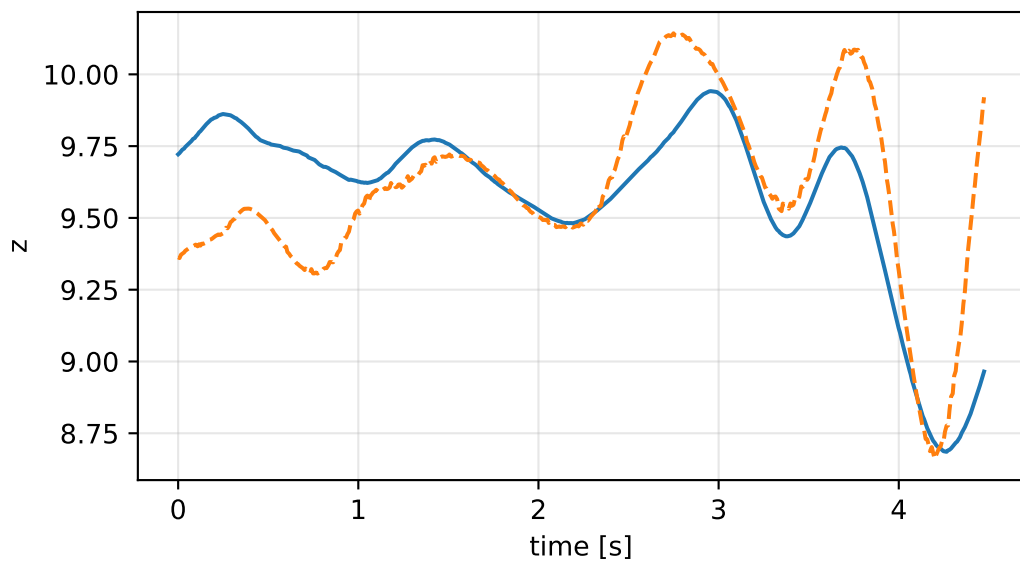
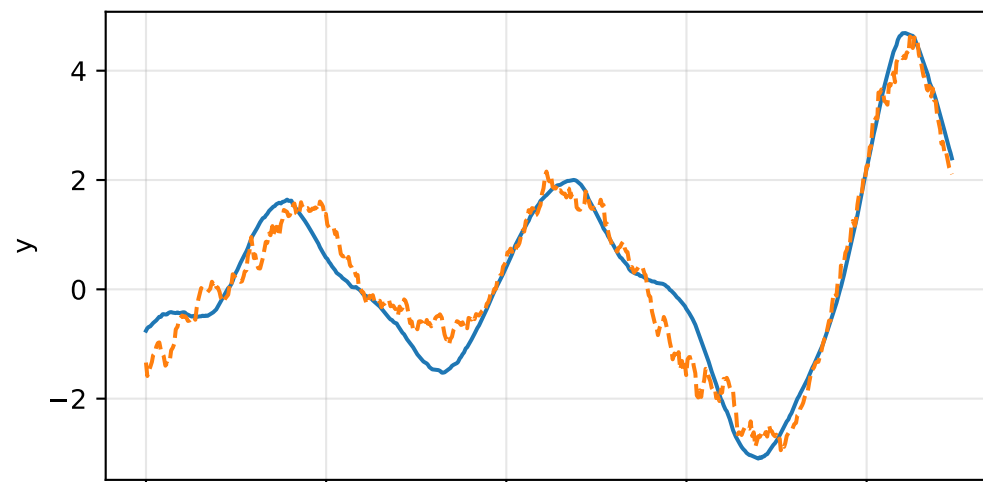
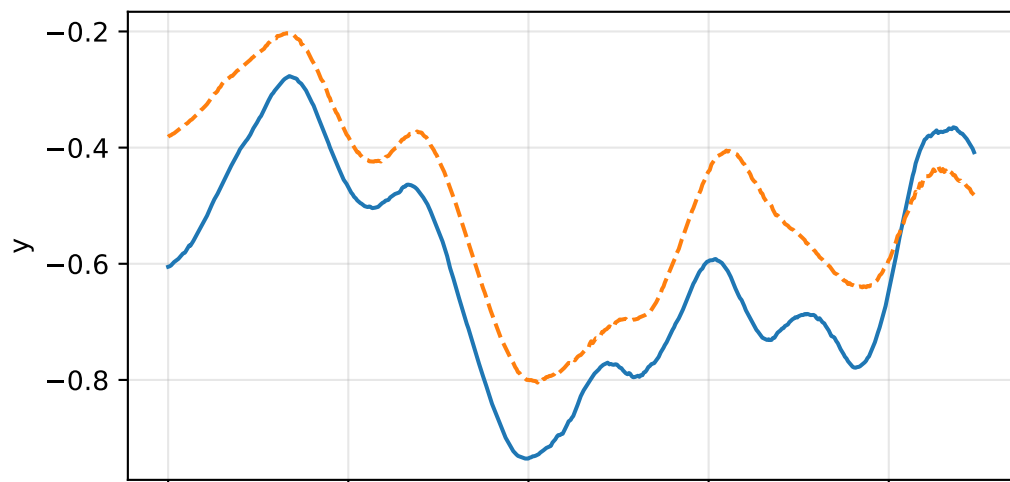
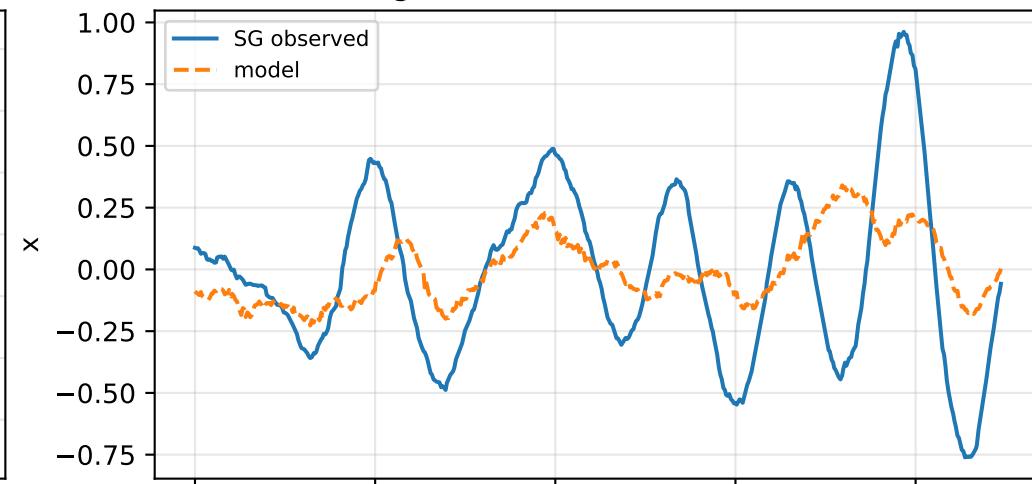


lag_legacy_smooth_schedule: SG trajectory and derived motion

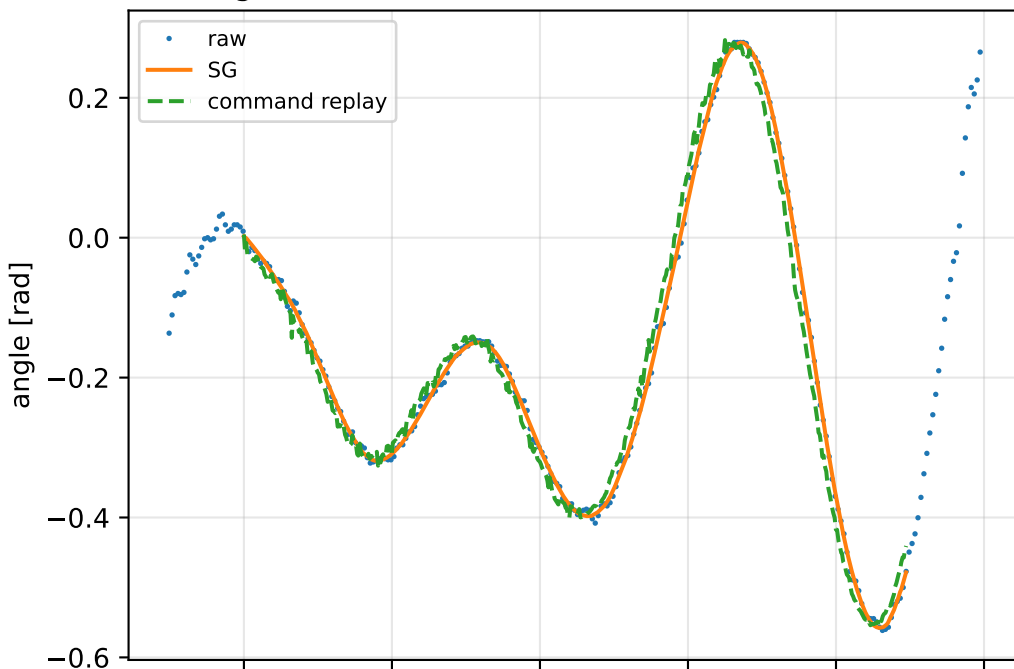


lag_legacy_smooth_schedule: acceleration residual objective

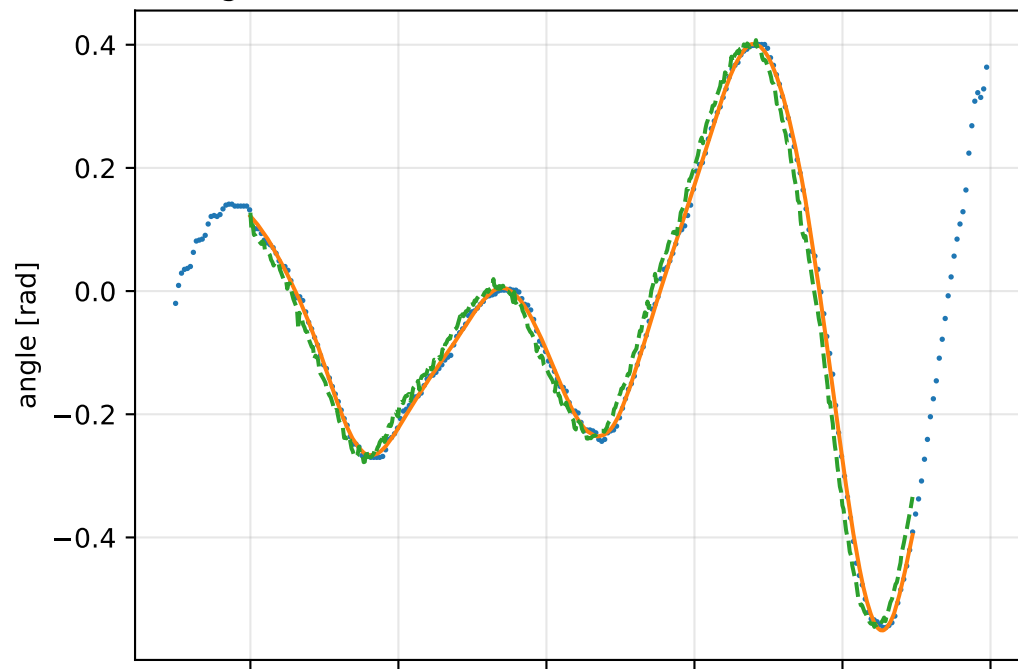
specific acceleration [m/s²]angular acceleration [rad/s²]

lag_legacy_smooth_schedule: gimbal measurement audit (objective=measured_sg)

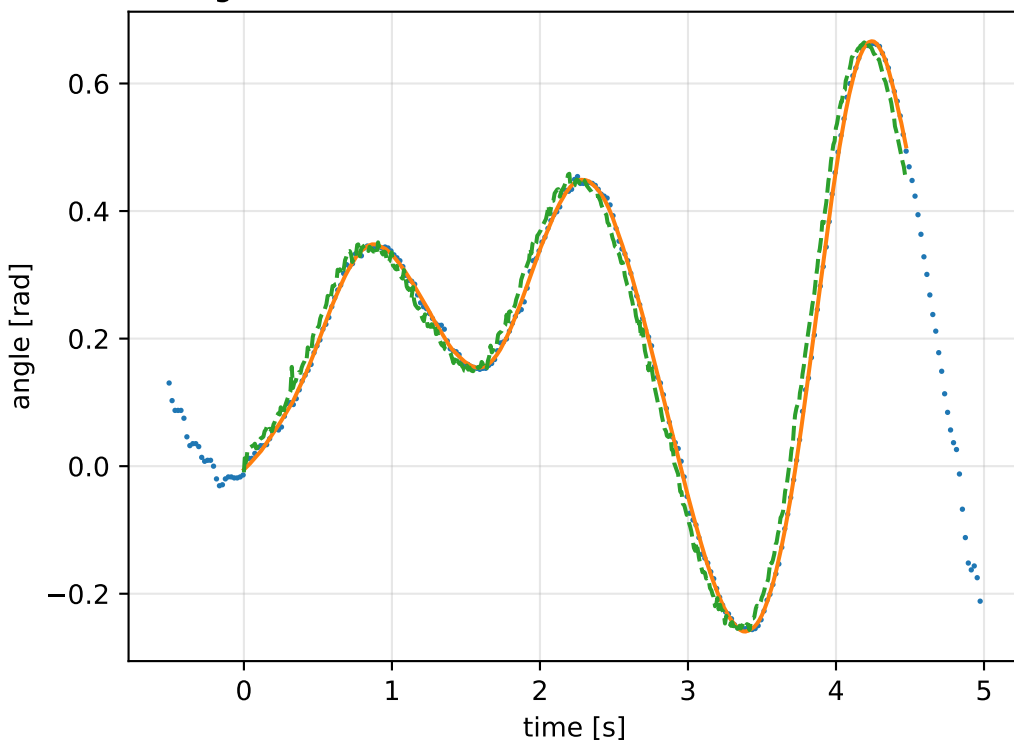
gimbal 1: RMSE=0.027, max=0.07529 rad



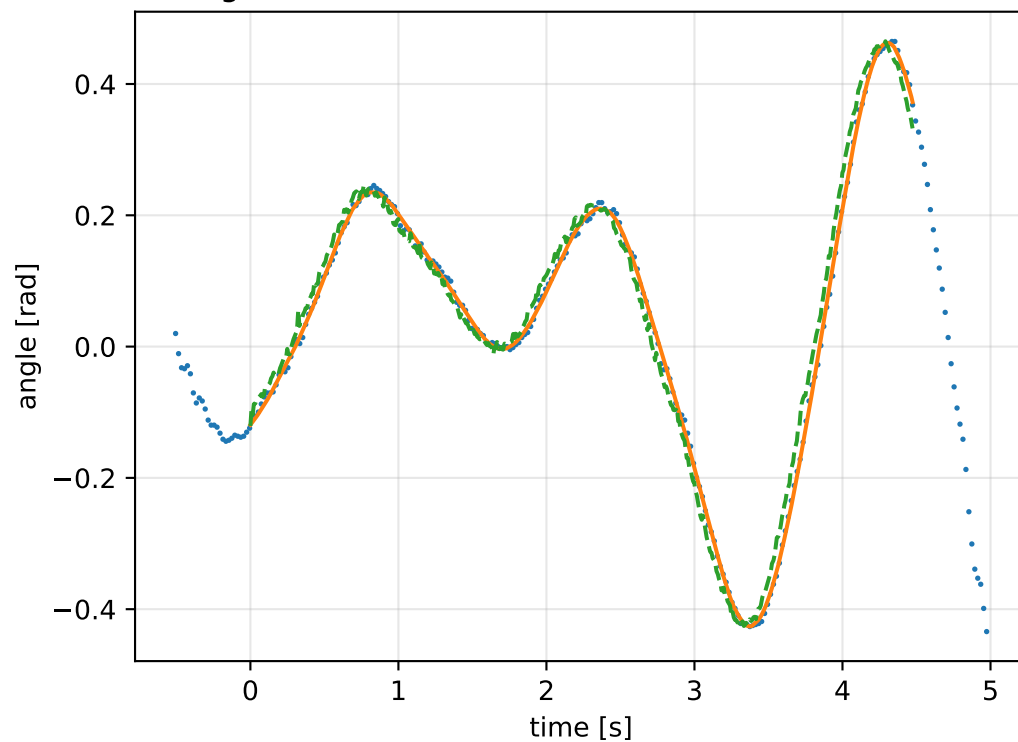
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

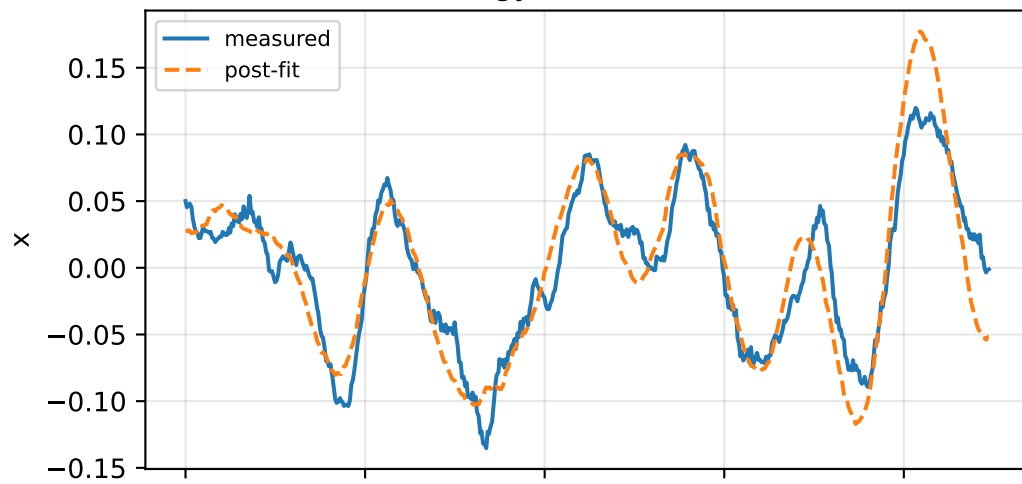
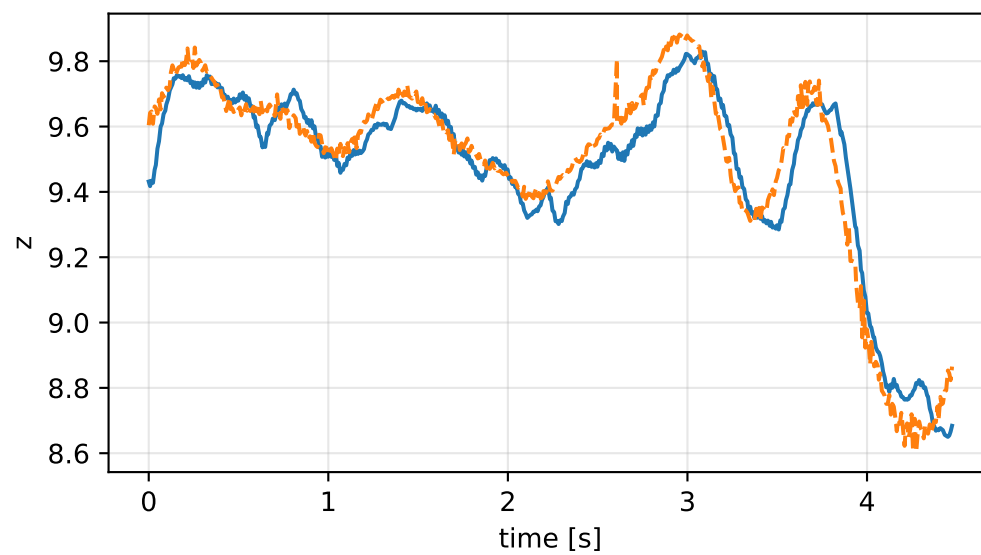
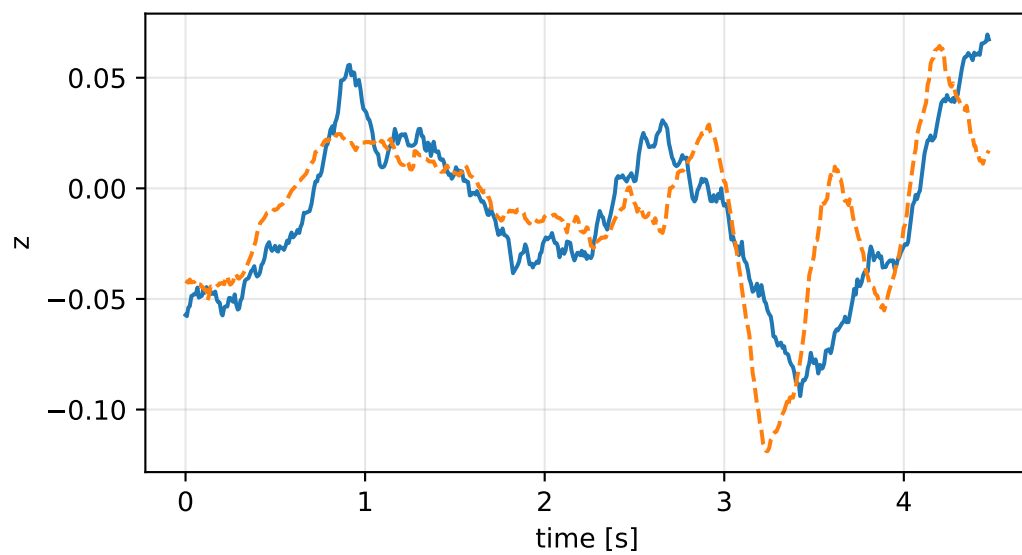
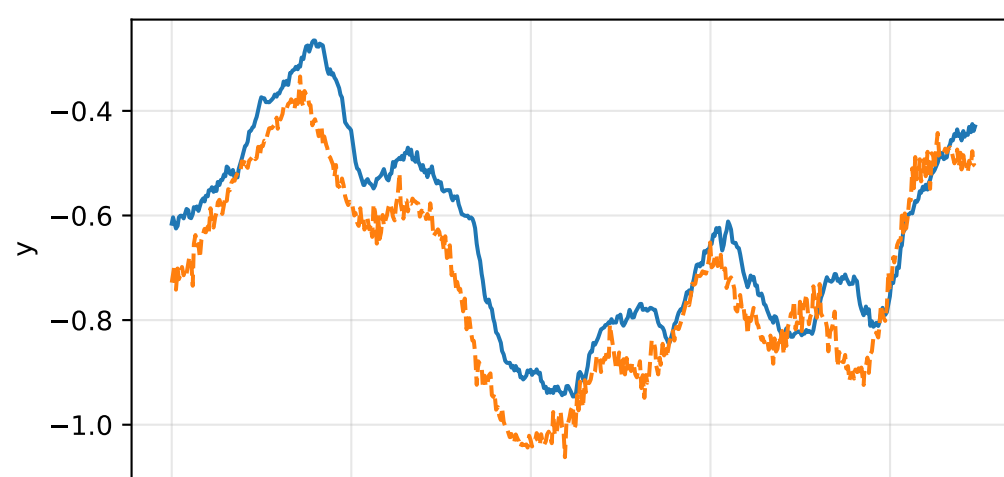
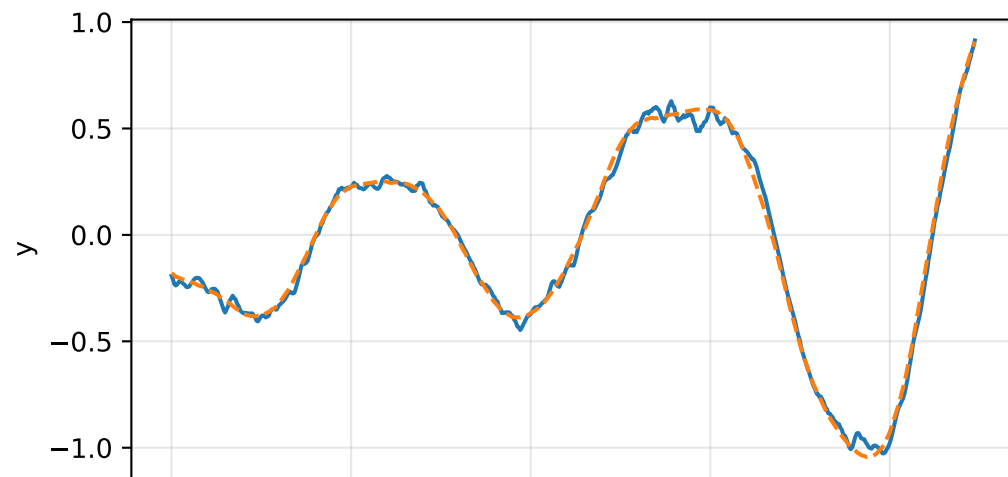
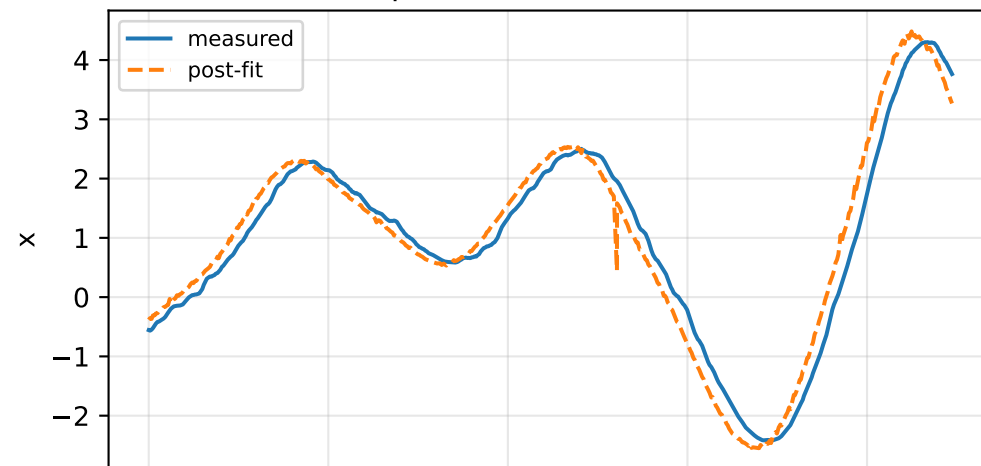


gimbal 4: RMSE=0.02598, max=0.07261 rad

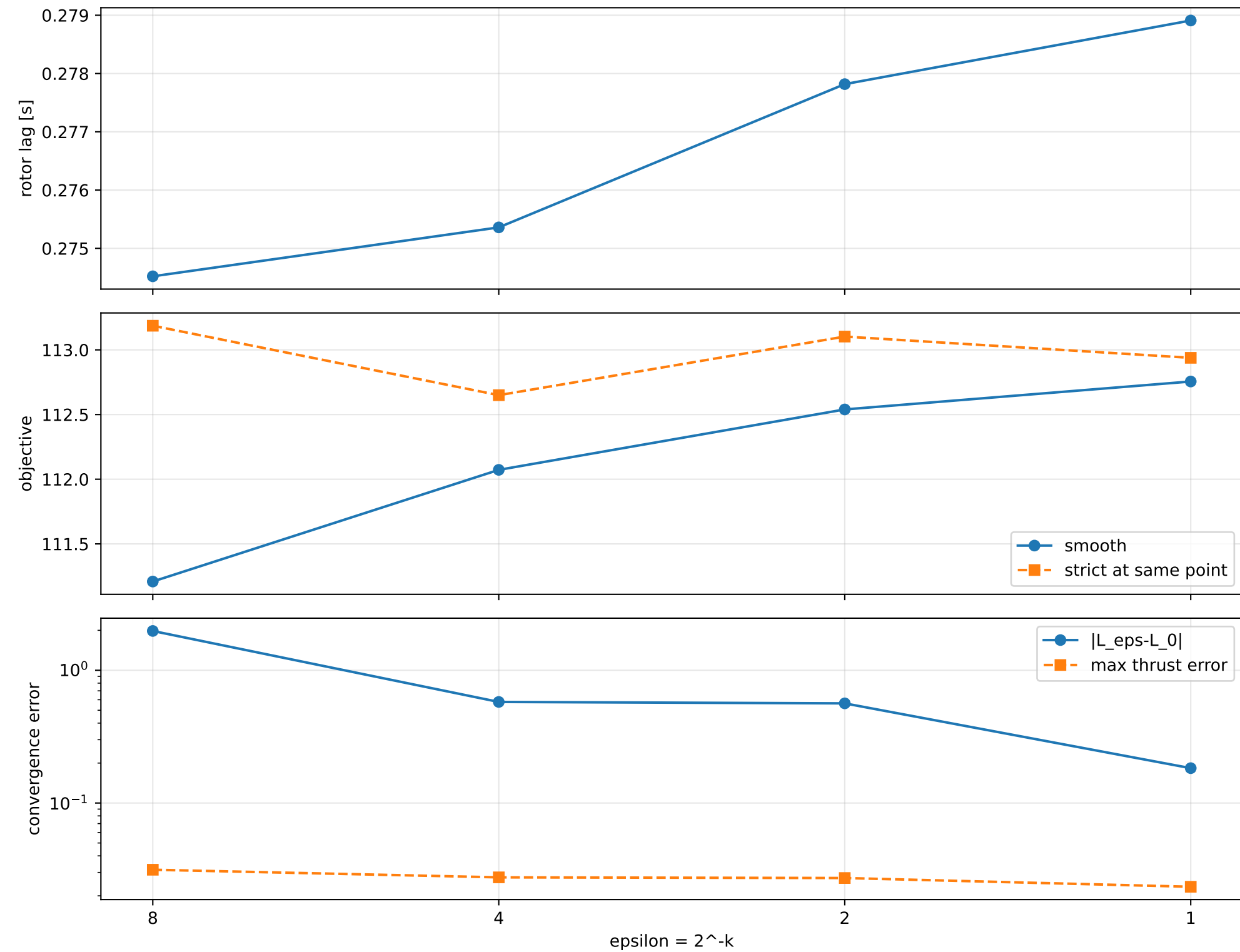


lag_legacy_smooth_schedule: IMU comparison (diagnostic only)

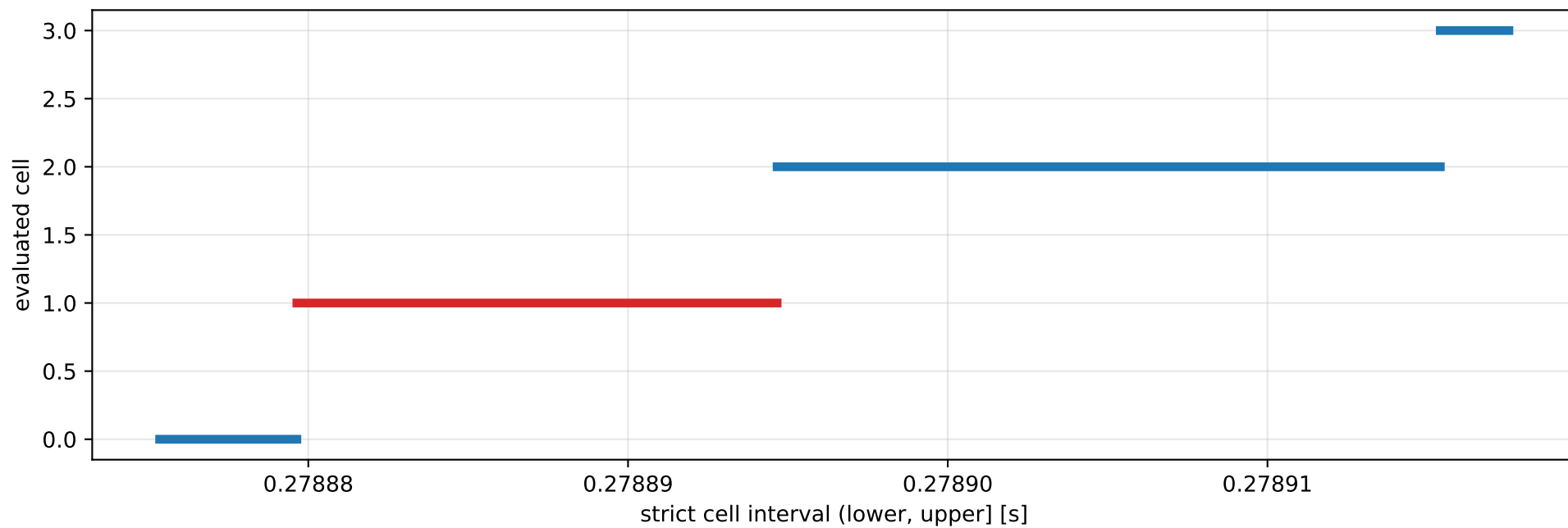
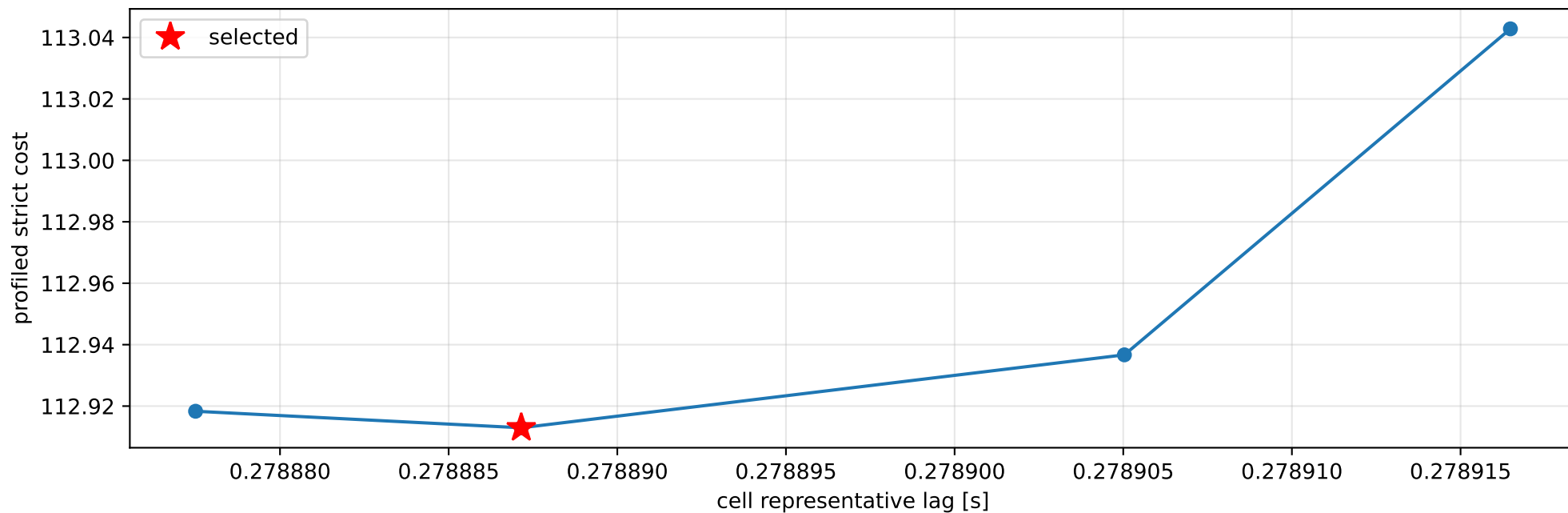
gyro [rad/s]

specific force [m/s²]

lag_legacy_smooth_schedule: smooth-to-strict continuation



lag_legacy_smooth_schedule: exact strict-ZOH lag cells



selected (0.278879642, 0.278894663] s; final smooth 0.278909658 s; data boundary=False

lag_legacy_smooth_schedule: parameter estimate

Case: lag_legacy_smooth_schedule

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.90982462

inertia [kg m²]:

```
[[ 4.13940062e-01  8.85855608e-03 -8.17294149e-05]
 [ 8.85855608e-03  1.66500236e-01 -5.77698152e-02]
 [-8.17294149e-05 -5.77698152e-02  5.63962063e-01]]
```

CoG body [m]: [-0.02830305 0.0125538 0.01007044]

force effectiveness: [1.88299608 1.48904744 1.67058453 2.03281964]

scale-free J/m [m²]:

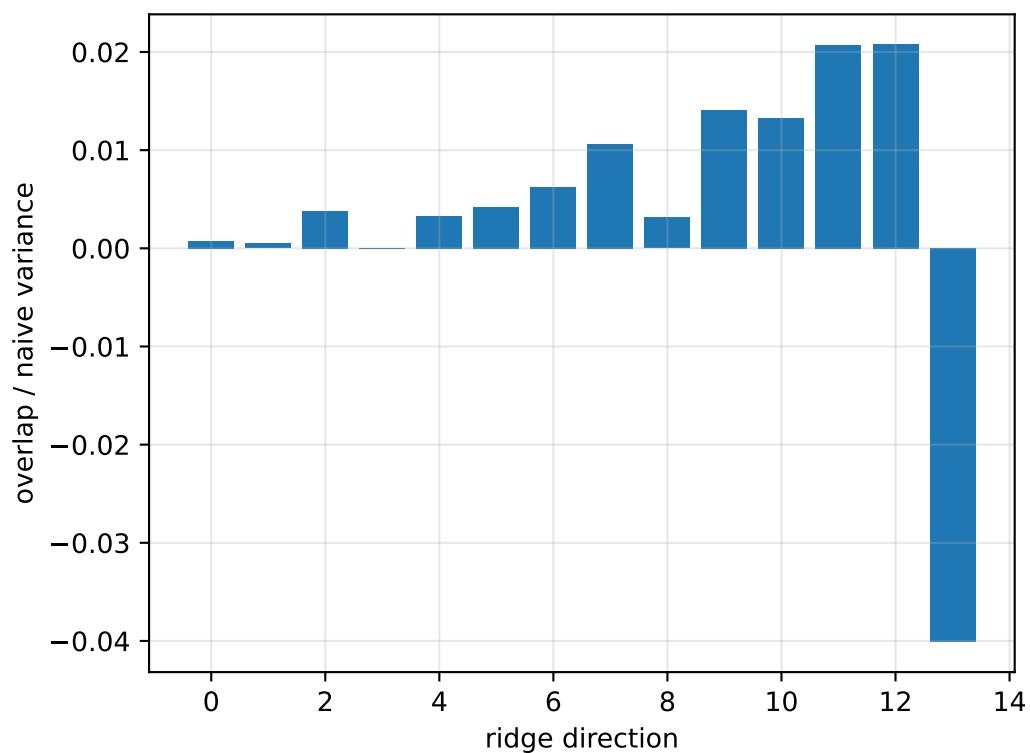
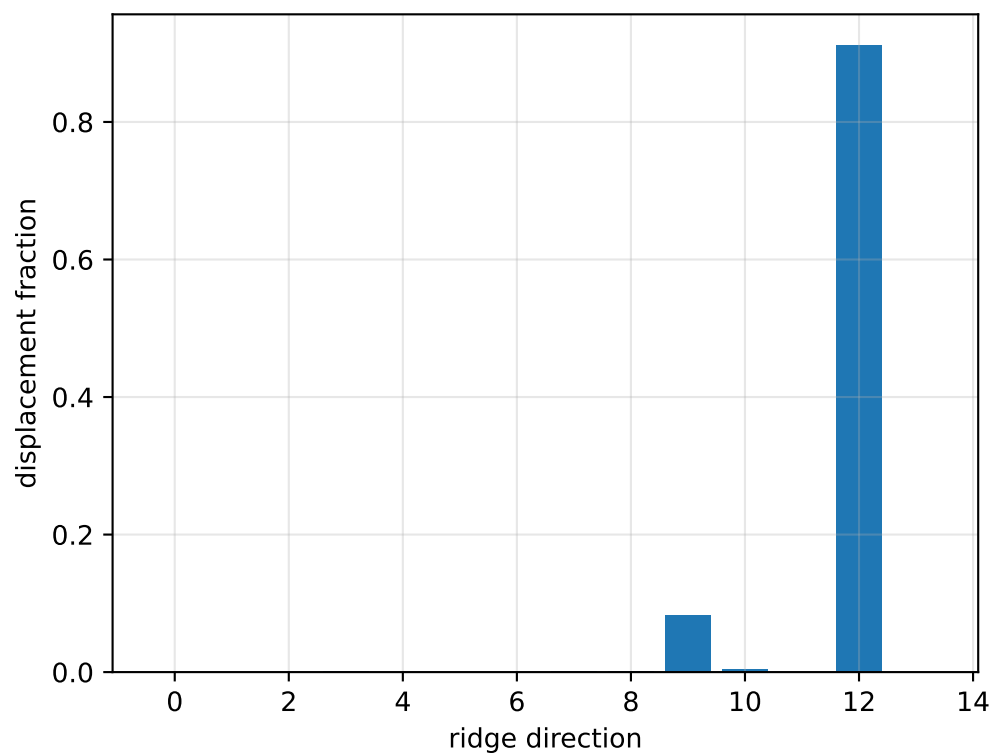
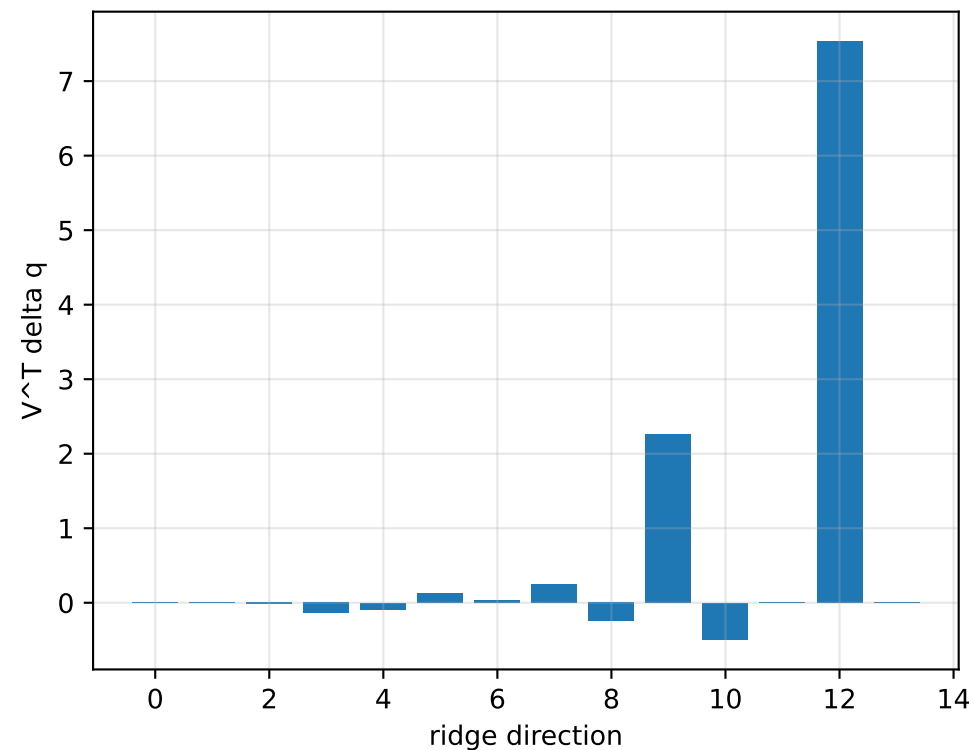
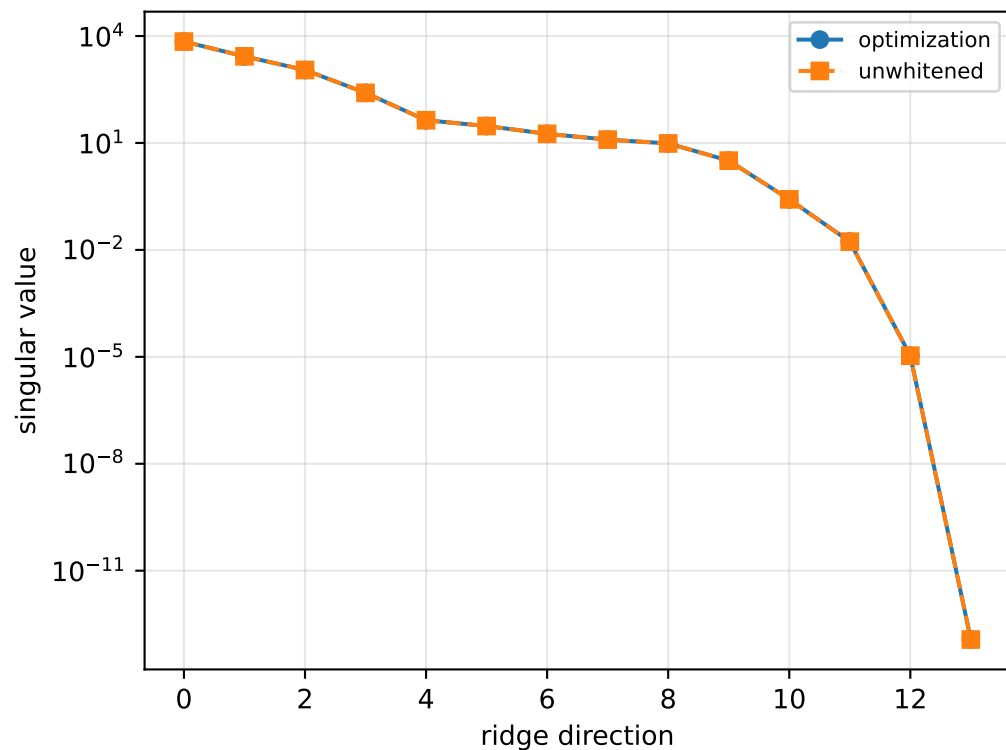
```
[[ 8.43085230e-02  1.80425102e-03 -1.66460966e-05]
 [ 1.80425102e-03  3.39116463e-02 -1.17661668e-02]
 [-1.66460966e-05 -1.17661668e-02  1.14863994e-01]]
```

scale-free f/m: [0.38351595 0.30327915 0.3402534 0.41403101]

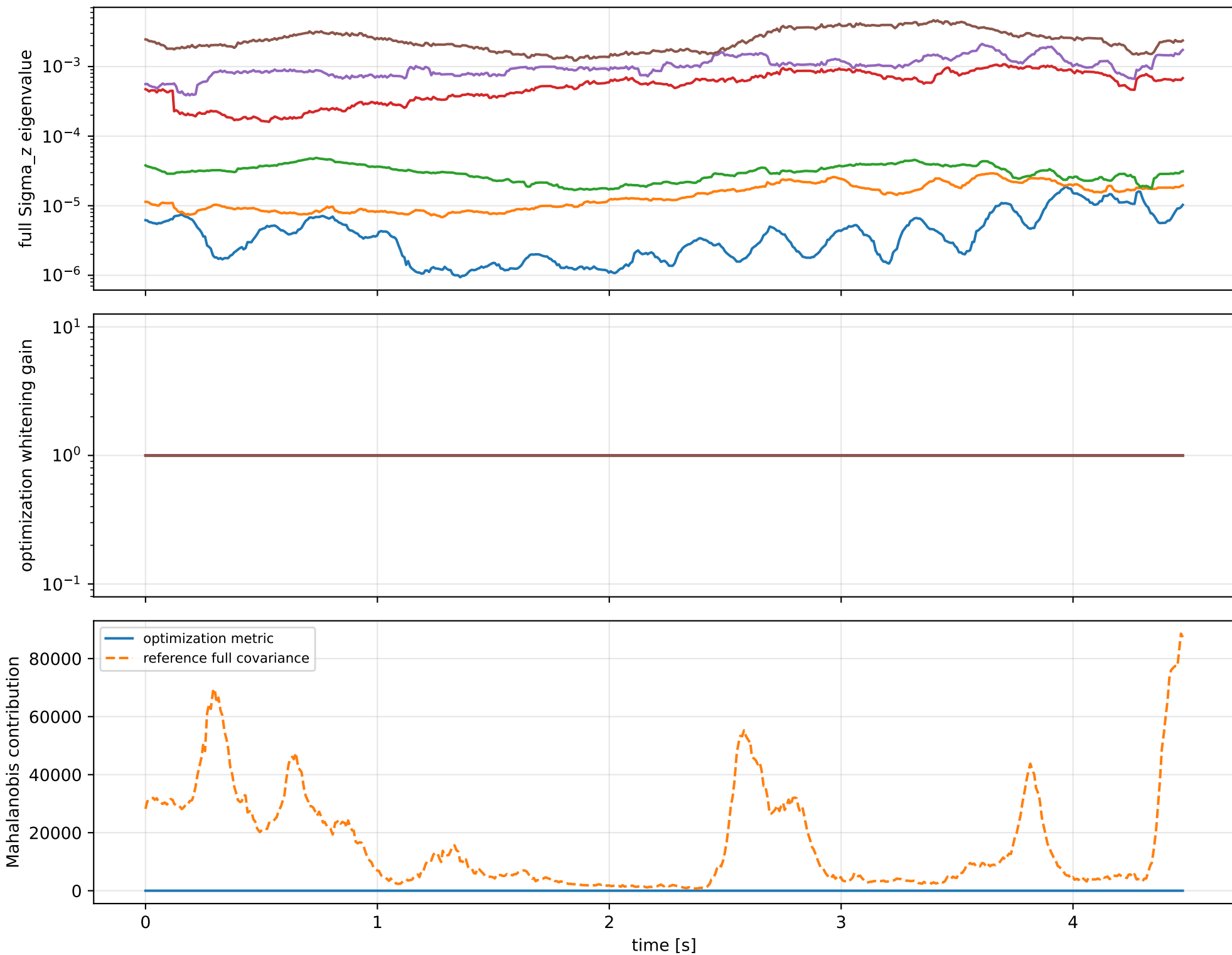
rotor lag cell: (0.278879642, 0.278894663] s

representative: 0.278887153 s

lag_legacy_smooth_schedule: ridge spectrum (rank 13, nullity 1, ||jv||=1.48e-13)



lag_legacy_smooth_schedule: optimization vs reference covariance



lag_legacy_smooth_schedule: wrench / closure (force max 1.33e-14, torque max 2.15e-16)

